

Solar Power Generation Forecasting Service

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Abstract— Starting in February 2019, the small-scale electricity brokerage market was opened and electricity aggregators were allowed to sell electricity markets. The ability of electricity aggregators to accurately predict the generation of electricity is competitive in electricity market and Renewable Energy Certificates (REC) transactions. Therefore, this paper presents a design and implementation case of the forecasting web service of the electricity generation from distributed energy sources for electricity aggregators of the small-scale electricity brokerage market.

Keywords— Solar, forecast, predict, electricity aggregators, web services

I. INTRODUCTION

Recently, Korea power exchange (KPX) has opened a small scale electricity brokerage market for electricity aggregators to collect and manage distributed energy resources [1].

Electricity aggregators were allowed to sell electricity generated from small scale distributed energy resources and renewable energy certificates (REC) in markets. This has the effect of mitigating the congestion in the distribution networks [2] caused by the surge of small scale distributed energy resources such as solar and wind power and increasing the profit of small-sized renewable energy generation companies. In addition, it is possible to collectively manage individual distributed energy resources through electricity aggregators, thereby enhancing the efficiency of power system operation, expecting forecast of generation in the power system and improving reliability and stability.

Small-scale electricity brokerage market is a form of submitting the amount of bid to the trading volume on the day before the trading day on the resource [3]. There is no an incentive policy to provide electricity aggregators for forecast accuracy yet. Currently, fourteen electricity aggregators are participating in the small-scale electricity brokerage market, but few are doing actual brokerage. Therefore, there is a need for an incentive policy to encourage market participation.

The government is currently discussing the provision of incentives for forecast of electricity generation from distributed energy sources to encourage small-scale electricity brokerage market participation.

If incentive policies for forecast of electricity generation from renewable energy are introduced, the ability of electricity aggregators to accurately predict the generation of electricity is competitive in electricity market and REC transactions.

In this paper, we implemented a web service of electricity aggregators for the forecast of electricity generation from distributed energy sources based on long short-term memory (LSTM) algorithm [4].

II. ELECTRICITY MARKET FOR SMALL DISTRIBUTED ENERGY RESOURCES AGGREGATOR

Electricity market in South Korea, only the centrally dispatched generation which capacity exceeds 20MW are participated in the bid in electricity market. Power plants less than 20 MW and including distributed energy resources, sell electricity at system marginal prices (SMP) to the electricity market without bidding.

Distributed energy resources such as solar and wind power is intermittently affected by the weather, and is unreliable because the output is variable and unpredictable. As the distributed energy resources increases, the stability of the electricity supply and demand will get worse year by year. Therefore, KPX as an electricity dispatch operation center need to manage and control of distributed energy resources.

Starting in February 2019, the small-scale electricity brokerage market was opened and electricity aggregators were allowed to sell markets. The small-scale electricity brokerage business is a business that sells electricity and renewable energy certificates (REC) by aggregating less than 1MW of distributed energy resources such as solar, wind and energy storage system (ESS). This has the effect of mitigating the power system congestion caused by the surge of distributed energy resources such as solar power and increasing the profit of small power generation companies. In recent years, various types of electricity brokerage business market is open in the US, Europe, and Japan.

Therefore, an energy management platform for electricity aggregators and renewable energy forecasting technology for the small-scale electricity brokerage market bidding is required. Energy Management platform for electricity aggregators have been studied in many researches [5], [6].

In the next section, we describe the implementation of web services of electricity aggregators for using the power generation forecast technique. Renewable energy in South Korea accounts for most of the solar power, it is limited to predict the electricity produced by solar power generation service in this paper.

III. SCENARIO OF SOLAR POWER FORECAST SERVICE

Figure 1 shows the composition of the forecasting web service for electricity aggregators of the small-scale electricity brokerage market. The scenario of web services can be composed of membership registration and login, introduction of power generation forecasting model, process for training deep learning model, day-ahead hourly forecasting of solar generation, statistical analysis, and management of deep learning model life cycle.

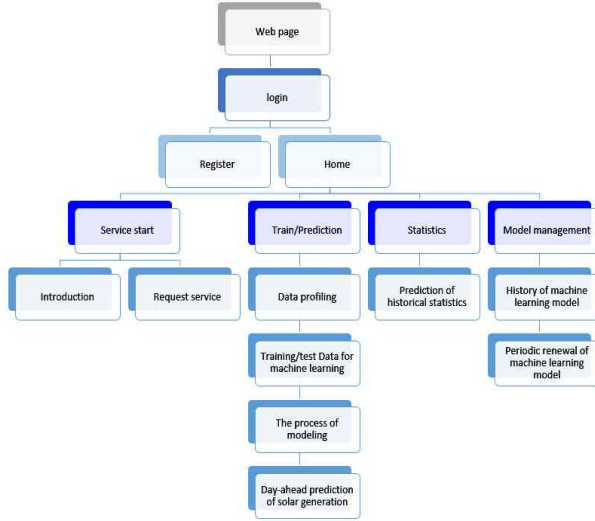


Figure 1. Flow diagram for forecast of solar generation

Input data for time series forecasting of the power generation deep learning model are as follows.

- Solar inverter data: power generation
- Environmental data: horizontal solar radiation
- Weather forecast data: current temperature, humidity, wind direction, wind speed, precipitation, sky status code, precipitation status code, precipitation probability

Electricity aggregators who have the historical power generation data can use the weather forecast data to create deep learning models. However, electricity aggregators that do not have the historical power generation data estimate the historical solar radiation data of the region where the solar power plant is installed and make deep learning models using weather data for the next day.

The power generation forecast model is constructed by using the LSTM algorithm specialized for the time sequence data to generate forecast model. It consist of several layers LSTM model, loss function and an optimization method.

Figure 2 shows a use case scenario for solar power forecasting service. First, input data (power generation, work love, weather information, etc.) is divided into train and test data sets for the forecast model generation.

Second, train the forecast model based on train data set. Third, the forecast of error rate and forecast accuracy of the learning data are evaluated. It determines the train status based on the loss and accuracy trends according to the number of repetitions. Fourthly, the model in which training ends is evaluated using test data set. Fifth, the final forecast model is stored. Sixth, forecast the solar generation by using the next day weather forecast information on the generated forecast model. Seventh, the results of power generation forecast for each resource are stored. The electricity aggregators check the day-ahead hourly forecasting of solar generation for each resource and use it as a full-trade bid.

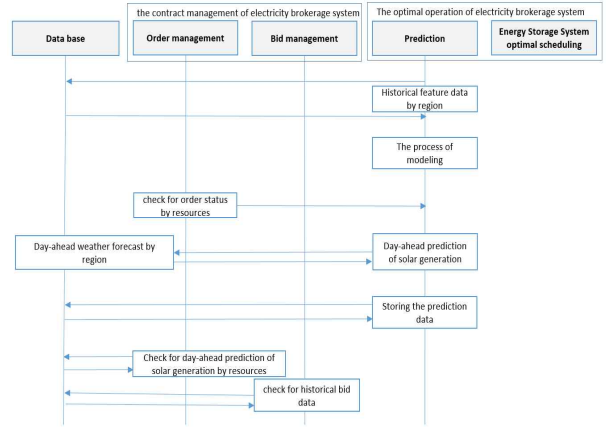


Figure 2. Use-case for forecast of solar generation

Solar power generation forecasting model was implemented based on tensorflow library in python, and web service for forecasting was implemented based on Java. Because the language between the web service language and the forecast model is different, we used a web server framework bottle and web socket server for python.

The bottle library is used for operations such as uploading input data set files to the python web server and training predictive models. The web socket server is used to transfer training completion messages between the train thread module of the forecast model and the web server thread.

A. Training process

Figures 3 shows the structure of the training module part of the power generation forecast model. The power generation forecast model training process is as follows. Through the power generation forecasting service web, the electricity aggregators can upload historical data information of the distributed energy resources to be predicted. The web service checks the integrity of the uploaded data and starts training the forecast model. After finishing the training, it notifies the electricity aggregators of completion of training, and stores the trained forecast model storage file path and distributed energy resource information in the database of power generation forecast service.

Figures 4 shows the forecast module part of the power generation forecast model.

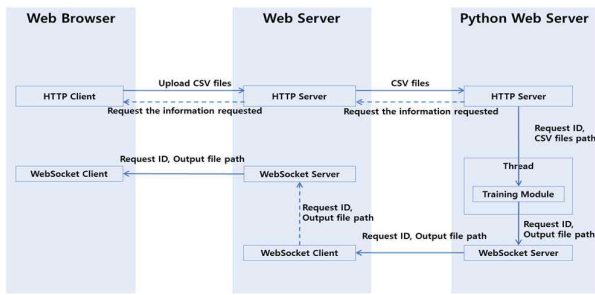


Figure 3. Training process for solar power generation forecast

B. Forecasting process

Figures 4 shows the forecast module part of the power generation forecast model. Once the forecast model training has been completed, the electricity aggregators can use the next day (day-ahead) of solar power generation forecasting service for the each distributed energy resources. The electricity aggregators select the resource to be used by the service in the web browser and executes the solar power generation forecast. The web server receives the next day weather forecast data based on the x, y coordinates of the region where the selected distributed energy resource is installed from the Korea Meteorological Administration (KMA) in the form of xml, and uses it as an input data of the forecast model. The forecast results are stored in a database and visualized with Web service.

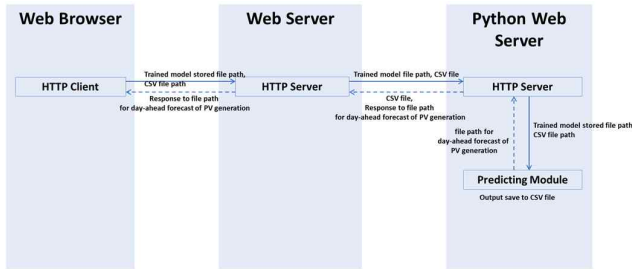


Figure 4. Solar forecast model forecasting

IV. RESULTS

Figure 5 shows the implementation of solar forecasting Web service for electricity aggregators. The left figure explains the LSTM model and parameters used in the forecasting process and forecasting. The right figure shows a screen for registering the resources recruited by the electricity aggregators and requesting the power generation forecast.

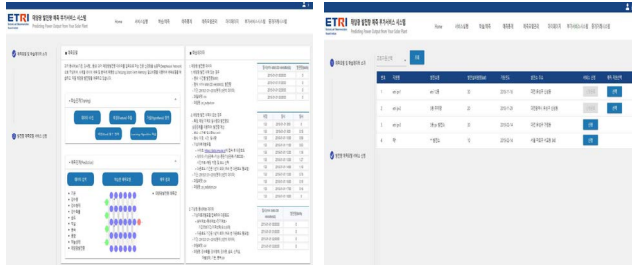


Figure 5. Solar power generation forecast web service

When the electricity aggregators register the address where the distributed energy resources are installed, it store the information converted into x and y coordinates, collects the data of the weather forecast data, and predicts the solar power generation.

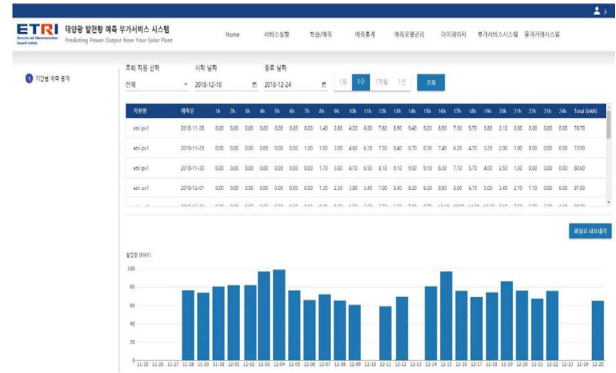


Figure 6. Statistical results

Figure 6 shows statistical information on the results of Solar power generation forecast. The solar power generation forecasting service also implemented a function to update the old data by setting the forecast model re-training.

V. FUTURE WORK

In this paper, we design and implement a technical structure for web services for solar power generation forecast. In the future, we plan to study how to predict the power generation of multiple solar power sites using a forecasting single model.

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